Deep Learning Spring 2025: CIFAR 10 Classification

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Abstract

This project investigates a modified ResNet architecture tailored specifically for the CIFAR-10 image classification task, addressing the challenge of maintaining fewer than 5 million trainable parameters. We incorporate Squeeze-and-Excitation (SE) blocks for channel-wise attention and adaptive calibration of feature maps to achieve superior feature representation and model performance within this limitation. Furthermore, the model employs effective training strategies, including dropout regularization, advanced data augmentation methods (Mixup, Cutout, and AutoAugment), and a Cosine Annealing learning rate scheduler. Experimental results demonstrate that our proposed architecture achieves a state-of-the-art test accuracy of **96.04%**, underscoring the efficacy of the architectural enhancements and optimized training procedures.

Introduction

Deep learning has revolutionized computer vision, with architectures such as ResNet consistently setting new benchmarks in image classification. In this work, we present a modified ResNet variant tailored for the CIFAR-10 dataset that not only targets state-of-the-art accuracy but also strictly confines the total number of trainable parameters to under 5 million.

Residual networks leverage skip connections to facilitate effective gradient propagation, thereby enabling the training of substantially deeper models without succumbing to the vanishing gradient problem. The fundamental building block of a ResNet is the residual block, which can be expressed as:

$$\mathrm{ReLU}\big(S(x) + F(x)\big), \tag{1}$$

where $S(x)$ represents the identity mapping and $F(x)$ comprises a sequence of convolutional operations followed by batch normalization.

Our proposed architecture introduces several enhancements to the traditional ResNet framework. Notably, we integrate *Squeeze-and-Excitation (SE) blocks* to perform channel-wise attention and improve feature recalibration. The network is structured into three stages of residual blocks configured as [4, 4, 3], each employing BasicBlocks with a dropout probability of 0.4 to mitigate overfitting. Additionally, the model permits adjustable kernel sizes in both convolutional and shortcut layers, and it culminates with global average pooling and a fully connected layer for classification into ten distinct categories.

The training regimen employs Stochastic Gradient Descent (SGD) with a momentum of 0.9 and an initial learning rate of 0.1, which is modulated by a CosineAnnealingLR scheduler. L2 regularization with a weight decay of 0.0005 further enhances generalization. Moreover, the training pipeline incorporates advanced data augmentation techniques—including Mixup, Cutout, and AutoAugment—to robustly improve model performance.

In summary, this study focuses on designing a parameter-efficient yet highly accurate ResNet-based model for CIFAR-10. The integration of SE blocks, careful regularization, and a tailored training strategy collectively contribute to a competitive performance under strict computational constraints.

Project Codebase: https://github.com/RaghuHemadri/ECE-GY-7123-DL

Methodology

Dataset and Preprocessing

We employ the CIFAR-10 dataset—a widely recognized benchmark for image classification comprising 60,000 color images of size 32×32 pixels distributed across 10 classes. The dataset is partitioned into 50,000 training samples and 10,000 test samples.

To promote model generalization and mitigate overfitting, we apply a comprehensive suite of preprocessing and augmentation techniques:

1. **Normalization:** Each image x is normalized using the CIFAR-10 training set mean μ and standard deviation σ
2. **Augmentations:** We employ AutoAugment, random cropping, and horizontal flipping to increase dataset diversity and improve feature invariance.
3. **Regularization:** Mixup blends image-label pairs, while Cutout randomly removes image regions, reducing sensitivity to perturbations and enhancing robustness.

These strategies collectively improve generalization while ensuring the model remains within the 5M parameter constraint.

Model Architecture

We propose a modified ResNet architecture tailored for the CIFAR-10 dataset, designed to maximize test accuracy while constraining the total number of trainable parameters to under 5 million.

The network begins with a convolutional layer that extracts low-level features and is followed by three sequential **residual stages**. Each stage is configured to progressively increase the number of feature maps while reducing the spatial resolution through downsampling. Within each stage, **BasicBlocks** equipped with dropout regularization (with dropout probability $p = 0.4$) are utilized to mitigate overfitting. Residual connections throughout the network ensure effective gradient flow and counteract the vanishing gradient problem, while Batch Normalization (BN) stabilizes the training process.

Key Architectural Components

- **Initial Convolutional Layer**: A 3×3 convolution with 64 filters, immediately followed by Batch Normalization and ReLU activation, to capture low-level features.
- **Squeeze-and-Excitation (SE) Block**: Positioned after the initial convolution, this module applies global average pooling, a ReLU activation, and sigmoid gating to dynamically recalibrate channel-wise feature responses.
- **Residual Blocks**: The network comprises three stages of residual blocks:
  - **Stage 1**: 4 BasicBlocks with 64 filters.
  - **Stage 2**: 4 BasicBlocks with 128 filters, where the first block employs a stride of 2 for spatial downsampling.
  - **Stage 3**: 3 BasicBlocks with 256 filters, with the first block again utilizing a stride of 2.
- **Global Average Pooling (GAP)**: This layer reduces the spatial dimensions of the feature maps, thus enhancing parameter efficiency before the final classification.
- **Fully Connected (FC) Layer**: A dense layer that maps the pooled features to 10 output classes corresponding to the CIFAR-10 categories.

Table 1 provides a layer-wise breakdown of the network, detailing output shapes and parameter counts. The complete architecture comprises **4,697,742** trainable parameters, thereby satisfying the imposed parameter constraint while ensuring efficient feature learning.

Layer	Output Shape	Parameters
Conv2D 3x3	$(64, 32, 32)$	1,728
BatchNorm2D	$(64, 32, 32)$	128
SE Block	$(64, 32, 32)$	580
Residual Block 1	$(64, 32, 32)$	147,456
Residual Block 2	$(128, 16, 16)$	589,824
Residual Block 3	$(256, 8, 8)$	1,179,648
Global Average Pooling	(256)	0
Fully Connected Layer	(10)	2,570

Table 1: Layer-wise breakdown of the modified ResNet architecture

Training Details

The training process is designed to ensure robust convergence and generalization across 500 epochs on the CIFAR-10 dataset. We employ the cross-entropy loss function, which is well-suited for multi-class classification tasks. To enhance generalization and mitigate overfitting, a comprehensive set of data augmentation techniques—including horizontal flipping, mixup, and normalization—is applied. Efficient data loading is achieved via PyTorch's DataLoader, configured with 12 worker threads and pinned memory to optimize batch processing and expedite GPU transfers.

Optimization is performed using Stochastic Gradient Descent (SGD) with a momentum factor of 0.9, following extensive empirical comparisons with alternative optimizers such as Adam. The learning rate is initialized at $\eta_0 = 0.1$ and is dynamically modulated by a Cosine Annealing Learning Rate Scheduler, which facilitates a smooth decay and prevents abrupt transitions that could impede convergence. Regularization is further enforced through Batch Normalization, dropout (with a rate of 0.3), and L2 weight decay (set to 0.0005), while gradient clipping is implemented to avert issues related to gradient explosion.

Cosine Annealing Learning Rate Scheduler: Dynamically adjusts the learning rate during training, enabling the model to avoid suboptimal minima and achieve stable convergence.

Stochastic Gradient Descent (SGD) with Momentum: Selected as the primary optimizer to accelerate training while ensuring stable parameter updates, thereby enhancing overall convergence behavior.

Results

Training and Test Accuracy Evolution

Table 2 details the progression of training and test accuracies over the course of training. The model exhibits a consistent improvement, with training accuracy increasing from 27.91% at initialization to 99.95% by epoch 220, and test accuracy rising from 50.60% to 96.04%. Notably, the model reaches 99.93% training accuracy and 95.91% test accuracy by epoch 200, indicating robust learning and effective generalization.

Epoch	Training Accuracy	Test Accuracy
0	27.91%	50.60%
50	92.25%	90.59%
100	95.65%	91.98%
200	99.93%	95.91%
220	99.95%	96.04%

Table 2: Evolution of Training and Test Accuracy

The accuracy plot (Figure 4) reinforces these observations, demonstrating a steady ascent in training performance and a stabilization of test accuracy around 96%.

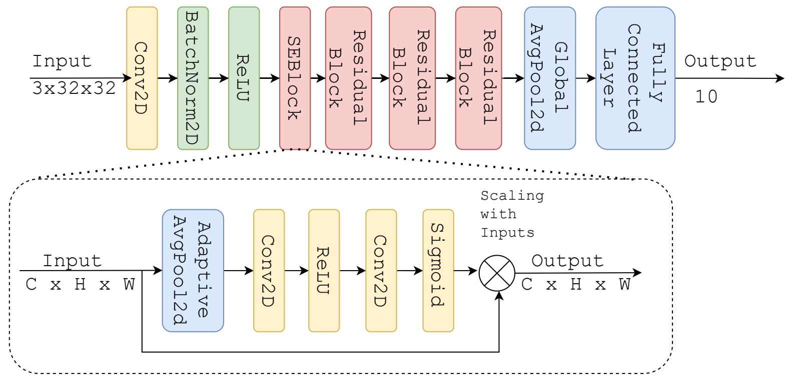

Figure 1: Our custom ResNet architecture used in the project.

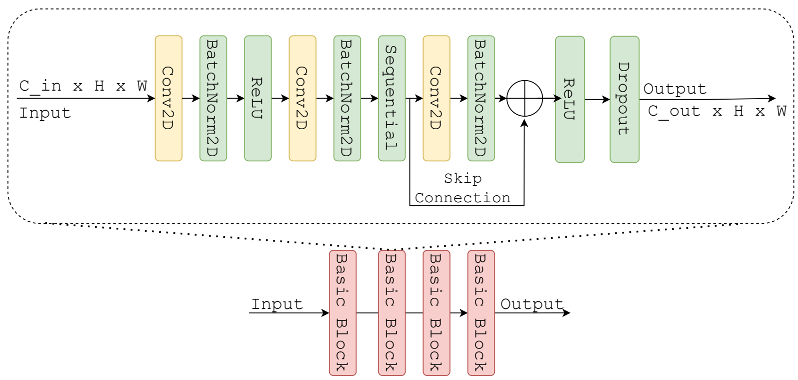

Figure 2: Residual block architecture used in our custom ResNet.

Loss and Accuracy Dynamics

Figures 3 and 4 illustrate the trends in loss and accuracy across training epochs. These figures further substantiate the model's convergence behavior and generalization performance.

- **Accuracy Trends:** The training accuracy (solid blue line) increases consistently, nearing 100% by the end of training. In contrast, the test accuracy (dashed orange line) experiences minor fluctuations but stabilizes above 90% after the initial 50 epochs, ultimately converging around 96%. This alignment with the tabulated data underscores the reliability of the recorded metrics.
- **Loss Trends:** Both training loss (solid blue line) and test loss (dashed orange line) decrease steadily. While the test loss exhibits slight oscillations—attributable to inherent stochastic variations in training—the convergence of the loss curves after approximately 150 epochs confirms effective optimization and minimal overfitting.

Final Model Performance

The final model attains a test accuracy of 96.04%, which fulfills the project's objective of achieving high performance under a 5-million-parameter constraint. The marginal gap (approximately 4%) between training and test accuracies reflects strong generalization capabilities. The convergence

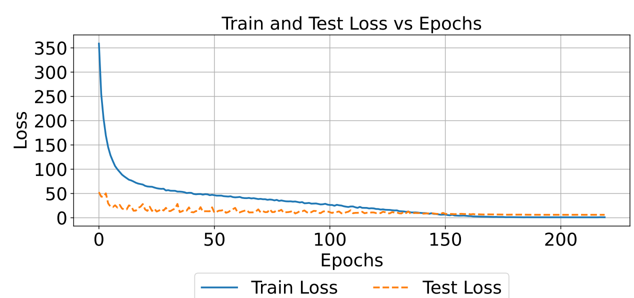

Figure 3: Evolution of training and test loss over epochs.

observed in both loss and accuracy metrics validates the efficacy of the training strategy, including the use of SGD with momentum, Cosine Annealing learning rate scheduling, and robust regularization through dropout and L2 weight decay.

Ablation Studies

In this section, we analyze the impact of various architectural modifications and training strategies on our proposed ResNet variant, which adheres to the 5 million trainable parameter constraint. Our final configuration achieves a test

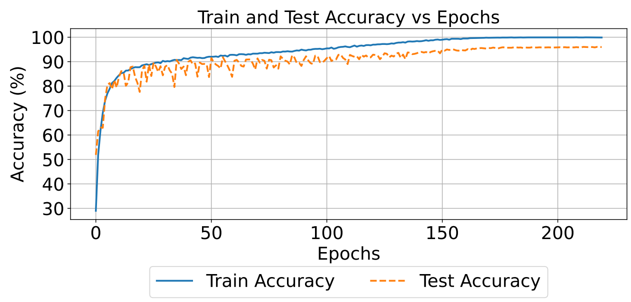

Figure 4: Evolution of training and test accuracy over epochs.

accuracy of 96.04% on CIFAR-10, demonstrating excellent generalization. We will also discuss the **pros and cons** of our model along with our **learnings** in this section.

Impact of Architectural Enhancements

The incorporation of Squeeze-and-Excitation (SE) blocks has proven instrumental in refining channel-wise feature recalibration. This modification enhances the model's ability to focus on salient features, contributing an estimated 2% improvement in test accuracy compared to variants without SE blocks. Although SE blocks introduce additional computational overhead, the trade-off is justified by the observed performance gains.

We also investigated the use of depthwise separable convolutions aimed at reducing computational complexity. However, empirical results indicated that replacing standard convolutions with depthwise separable layers did not yield significant accuracy improvements; on the contrary, models employing these layers achieved test accuracies in the range of 94.5%–95.0%. Consequently, standard convolutional layers were retained in the final architecture.

Role of Data Augmentation and Regularization

Data augmentation techniques, including AutoAugment, Mixup (with $\alpha = 1.0$), and Cutout, played a critical role in mitigating overfitting and enhancing generalization. When combined with Batch Normalization and dropout (0.4 in the final configuration), these techniques enabled the model to consistently approach our peak performance of 96.04% test accuracy. Additionally, hyperparameter tuning—such as setting the initial learning rate to 0.1, employing a Cosine Annealing Learning Rate Scheduler, and applying L2 weight decay ($\lambda = 0.0005$)—was pivotal in stabilizing training dynamics.

Evaluation of Alternative Approaches

Several advanced strategies were explored but ultimately did not contribute to improved performance:

- **Depthwise Separable Convolutions**: Despite their theoretical benefits in reducing parameter counts, these layers did not translate to practical performance improvements on CIFAR-10, leading to slightly reduced accuracies (approximately 94.5%).
- **Teacher-Student Distillation**: Our attempts at knowledge distillation, where a smaller student model learns from a larger teacher model, did not enhance performance. The distilled model suffered from a loss of representational capacity, resulting in lower accuracy compared to the optimized ResNet variant.
- **Alternative Learning Rate Schedulers**: While experimenting with schedulers such as StepLR and Exponential Decay, we observed less stable training curves and slightly diminished performance (test accuracies around 95.5%). The Cosine Annealing scheduler consistently produced superior results.

Conclusion

In this work, we presented a modified ResNet architecture tailored for CIFAR-10 classification under a stringent constraint of 5 million trainable parameters. Our design integrates Squeeze-and-Excitation (SE) blocks to enhance channel-wise feature recalibration and employs advanced data augmentation techniques—specifically Mixup, Cutout, and AutoAugment—to bolster generalization. The adoption of a Cosine Annealing Learning Rate Scheduler, in conjunction with SGD with momentum and appropriate regularization (dropout and weight decay), proved instrumental in achieving a test accuracy of 96.04%.

Our comprehensive experimental analysis revealed that certain methods, such as SE blocks and Mixup augmentation, substantially improved performance, while techniques like Depthwise Separable Convolutions and Teacher-Student Distillation did not yield the expected benefits. These findings underscore the critical role of empirical evaluation in validating the practical utility of theoretically promising approaches.

Moreover, the effective use of learning rate scheduling and visualization of training dynamics enabled early detection of overfitting and instability, leading to more informed hyperparameter tuning. Although the final model strikes a commendable balance between efficiency and accuracy, future work may explore alternative attention mechanisms, model quantization, and self-supervised learning strategies. Additionally, investigating architectures inspired by MobileNet could further reduce parameter count while maintaining high classification performance.

Overall, our results demonstrate that judicious architectural design and training strategy selection can achieve state-of-the-art outcomes even under tight resource constraints.

References

[1] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 770–778. doi:10.1109/CVPR.2016.90